

Revisiting the Intracranial Neurological Imaging Manifestations of Human Immunodeficiency Virus: Pictorial Review with a Pattern-Based Approach

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Abstract

Human immunodeficiency virus (HIV) leads to a spectrum of pathologies from head to toe. This review focuses on the neurological complications associated with HIV infection. It highlights both direct effects, resulting from virus' direct impact on central nervous system (CNS), and indirect effects, stemming from progressive decline in CD4 counts. Direct effect encompasses HIV-associated neurocognitive disorder (HAND) and HIV-associated vasculopathy. Indirect effect comprises all the opportunistic infections that an individual is predisposed too, including toxoplasmosis, cryptococcal meningitis, and progressive multifocal leukoencephalopathy. Radiological imaging plays a crucial role in diagnosis and distinguishing of all these entities, which can have overlapped clinical presentation. The imaging characteristics are distinctive, such as, HAND presents with symmetrical bilateral white matter hyperintensities on magnetic resonance imaging, while cerebral toxoplasmosis typically shows a target sign with multiple ring enhancement, and cryptococcal meningitis has a typical “soap bubble appearance.” Meanwhile, progressive multifocal leukoencephalopathy is identified by asymmetric T2 hyperintensities without enhancement. This review also covers CNS-immune reconstitution inflammatory syndrome, a critical condition linked with antiretroviral therapy. This broad spectrum of presentation underscores the need for a systematic, pattern-based approach to enhance diagnostic accuracy, which is provided at the end.

Keywords

- HIV
- MRI
- toxoplasmosis
- PML
- encephalopathy
- lymphoma
- cryptococcoma

Introduction

Human immunodeficiency virus (HIV) infection has widespread neurological implications, and leads to a spectrum of central nervous system (CNS) involvement. HIV-related neurological symptoms can be divided into two categories: (1) direct effects of HIV virus on the white matter and resulting

encephalopathy and vasculopathy causing clinical deterioration, and (2) indirect effects of HIV on the CD4 counts leading to susceptibility to opportunistic infections, namely, toxoplasmosis (CNS-Toxo), JC virus infection resulting in progressive multifocal leukoencephalopathy (PML), fungal infections such as cryptococcosis (CNS-Crypt), and of course infection with various strains of *Mycobacterium tuberculosis* resulting in tuberculosis (CNS-TB), and development of primary CNS lymphoma (PCNSL; ► **Table 1**).¹ Lastly, neurological complication resulting from antiretroviral therapy (ART), immune

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Table 1 Correlation between serum CD4-positive T lymphocyte counts and risk of HIV-related CNS diseases¹

CD4 count (cell/mm ³)	Frequently occurring CNS diseases
<50	Cytomegalovirus retinitis, primary CNS lymphoma, IRIS
<200	HIV encephalopathy, toxoplasmosis, cryptococcosis, PML
200–500	HIV encephalopathy
>500	Tubercular infection

Abbreviations: CNS, central nervous system; HIV, human immunodeficiency virus; IRIS, immune reconstitution inflammatory syndrome; PML, progressive multifocal leukoencephalopathy.

reconstitution inflammatory syndrome (IRIS), also accounts for a significant portion of neurological complication.

Despite being defined as distinct entities with an independent prognosis and specific management protocols, overlapping clinical features often complicate accurate diagnosis.

Thus, reaching a diagnosis with certainty can be an uphill task. Radiological imaging plays a pivotal role in diagnosing these conditions. By consolidating current knowledge and recent advancements in the field, this review underscores the crucial role of radiological imaging in differentiating these conditions; thus, bridging the gap between clinical symptomatology and definitive diagnosis. The systematic pattern-based approach aims to empower clinicians and radiologists to achieve early identification and appropriate management of HIV-related intracranial abnormalities, ultimately improving patient outcomes and enriching the scientific dialogue on neuro-HIV complication.

Direct Involvement by HIV Virus

HIV-Associated Neurocognitive Disorders (HAND)

Pathophysiology

The direct invasion of HIV into the CNS happens early in the disease, the virus persisting despite control of systemic viremia through a combination ART. This happens because many anti-HIV medications do not penetrate the blood-brain barrier, and HIV continues replicating in the brain.² The infection thus becomes chronic, inciting inflammation, neuronal injury, and synaptic dysfunction. Clinically no signs of cognitive impairment are present in the early stages of infection; however, diffuse myelin and axonal degradation sets in later and presents with cognitive symptoms termed HIV-associated neurocognitive disorders (HAND) in 30 to 50% patients. HIV-associated dementia is severe but milder neurocognitive impairment without dementia is often present. There are criteria specifying that two cognitive domains must be affected before labeling the HIV patient as HAND-positive.³

Imaging: Imaging shows symmetric bilateral T2 and FLAIR (fluid-attenuated inversion recovery) hyperintensities in the periventricular and deep white matter, and

rarely in the subcortical white matter and the brainstem, along with diffuse cortical atrophy (►Fig. 1).⁴ Advanced imaging techniques, such as magnetic resonance spectroscopy (MRS) and diffusion tensor imaging (DTI), can assess the early changes of neuronal damage and also subsequent response to therapy.⁵ MRS reveals lower N-acetyl aspartate (NAA) and higher myoinositol (ml) and choline (Cho) ratios, indicating neuronal loss and gliosis. DTI demonstrates an increase in mean diffusivity (MD) values and a decrease in fraction anisotropy (FA) values even when structural changes are not visible in the white matter on conventional MR sequences.

HIV-Associated Vasculopathy

Pathophysiology: HIV-associated inflammatory response and endothelial injury lead to vasculitis, stroke, and aneurysms/dissection.

Imaging: HIV vasculopathy mostly affects the small- to medium-sized arteries in the CNS. In young individuals, it typically causes multiple fusiform aneurysms and infarctions from cerebral artery stenosis or occlusion. Cerebral angiitis resulting from vessel wall inflammation manifests on magnetic resonance (MR) as thickening and enhancement of the vessel wall on vessel wall imaging.⁶ Computed tomography (CT) angiography and MR angiography are quite helpful in diagnosing this vascular manifestation of CNS-HIV. A representative case is shown in ►Fig. 2.

Indirect involvement by HIV

Cerebral Toxoplasmosis

Pathophysiology: Reactivation of latent *Toxoplasma gondii* leads to focal necrotizing encephalitis in immunocompromised patients. Presence of clinical symptoms, the identification of one or more mass lesions on imaging, and detection of the parasite in clinical samples are necessary for a conclusive diagnosis of CNS-toxoplasmosis.⁷

Imaging: Lesions of toxoplasmosis appear hypodense on CT, are frequently accompanied by intense surrounding vasogenic edema, and demonstrate ring-like enhancement. Although isolated lesions are observed in 15 to 20% of cases,⁴ multiple ring-enhancing lesions with adjacent vasogenic edema are the typical MR imaging (MRI) findings of CNS-toxoplasmosis (►Fig. 3). Thalamus, cerebellum, subcortical white matter, and basal ganglia are the most often affected areas.⁷ If the number of CD4-positive T cells in the serum is less than 50 cells/mm³, contrast enhancement might not be visible.

Pathognomonic imaging finding including target sign on T2-weighted images is characterized by concentric high- and low-signal areas with eccentric nodular target-like enhancement on contrast-enhanced T1-weighted imaging (T1WI) images.⁸ The eccentric enhancing nodule represents a cluster of thickened vessels, whereas the ring represents the inflammatory vascular zone at the margin of the lesion, which has a necrotic center.⁹ On susceptibility-weighted MRI, intraleSIONAL susceptibility signals, indicating areas of hemorrhages, can be seen in majority of cases (►Fig. 3D).

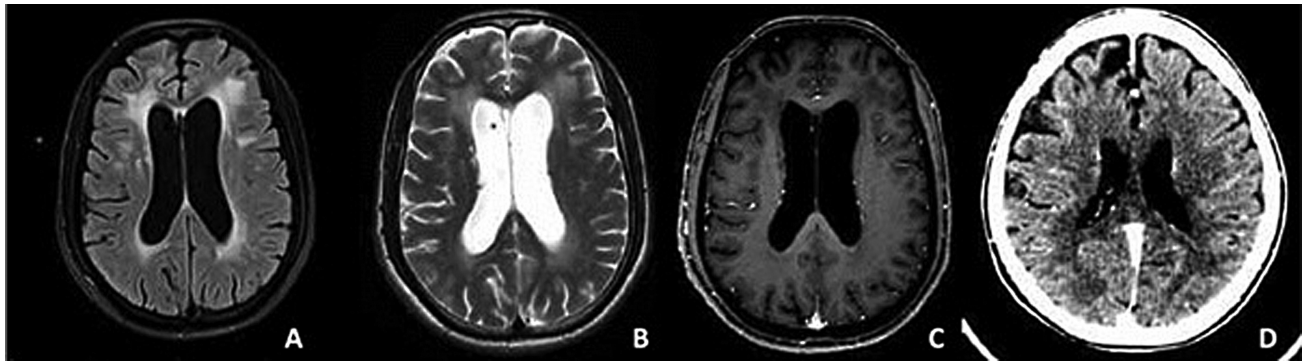


Fig. 1 In a 50-year-old male with HIV-positive status, (A) FLAIR and (B) T2WI axial images of brain showing bilateral symmetrical periventricular white matter hyperintensity extending into deep white matter and (C) post-contrast T1W FS axial images do not show any enhancement. (D) In a 40-year male HIV+ patient, axial CECT images show non-enhancing bilateral symmetrical periventricular white matter hypodensity extending into deep white matter with mild global cortical atrophy. Diagnosis: HIV-associated neurocognitive disorder. CECT, contrast-enhanced computed tomography; FLAIR, fluid-attenuated inversion recovery; HIV, human immunodeficiency virus; T1W FS, T1-weighted fat-saturated; T2WI, T2-weighted imaging.

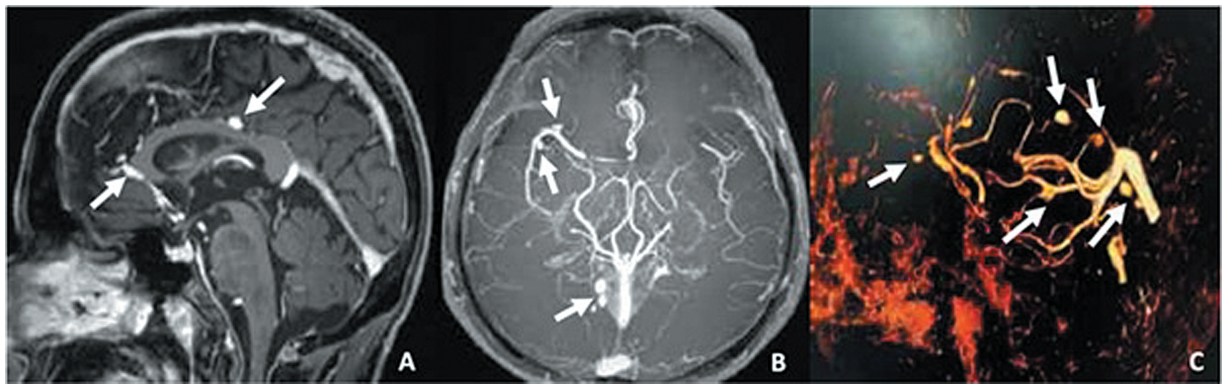


Fig. 2 In a 40-year-old male with HIV-positive status, (A) axial post-contrast T1FS, (B) MIP images, and (C) virtual rendered images show multiple saccular and fusiform outpouchings seen arising from right A2 and A3 segments of ACA, M2 segment of right MCA, and P4 segment of right PCA (arrows). Diagnosis: HIV-associated vasculopathy. ACA, anterior cerebral artery; HIV, human immunodeficiency virus; MCA, middle cerebral artery; MIP, maximum intensity projection; PCA, posterior cerebral artery; T1FS, T1-weighted fat-saturated.

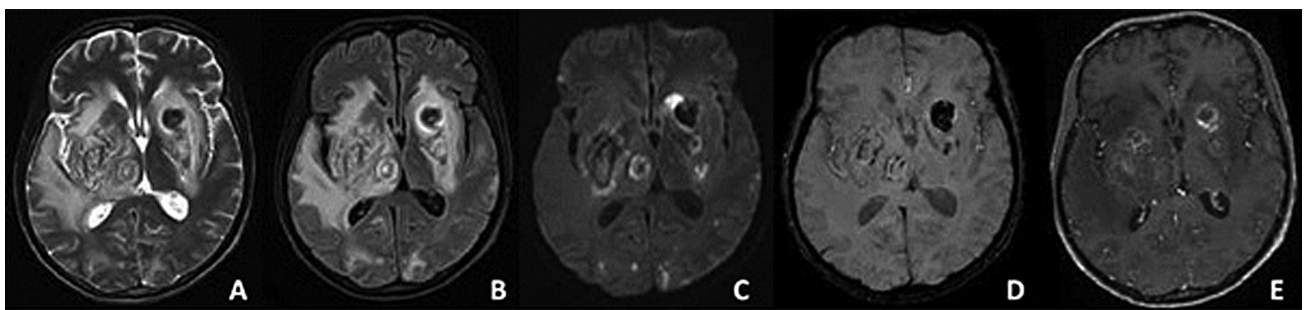


Fig. 3 In a 27-year-old male with HIV-positive status, (A) T2WI and (B) FLAIR images show multiple target lesion with alternating concentric areas of hyper- and hypointensity predominantly in bilateral basal ganglia and right thalamus with adjacent extensive perilesional vasogenic edema. (C) Axial DWI images show diffusion restriction. (D) SWI images show intralésional blooming corresponding with hemorrhagic foci and (E) post-contrast T1FS axial images show multiple ring-enhancing lesions in bilateral basal ganglia, right thalamus, and subcortical white matter of bilateral parieto-occipital lobe. The lesion in the left putamen shows eccentric nodule. Diagnosis: Cerebral toxoplasmosis. DWI, diffusion-weighted imaging; FLAIR, fluid-attenuated inversion recovery; HIV, human immunodeficiency virus; SWI, susceptibility-weighted imaging; T2WI, T2-weighted imaging.

Primary CNS Lymphoma

Pathophysiology: In HIV patients a high-grade B cell lymphoma associated with Epstein–Barr virus may occur. The incidence of PCNSL in HIV-infected individuals is 2 to 6%, almost 1,000 times higher than that in the general population.¹⁰

Although ART has decreased the incidence of HIV-related PCNSL, it remains the most common HIV-associated malignancy.¹¹ Unless treated, the median survival of HIV-infected patients with PCNSL, after the onset of clinical symptoms, is 1 month.

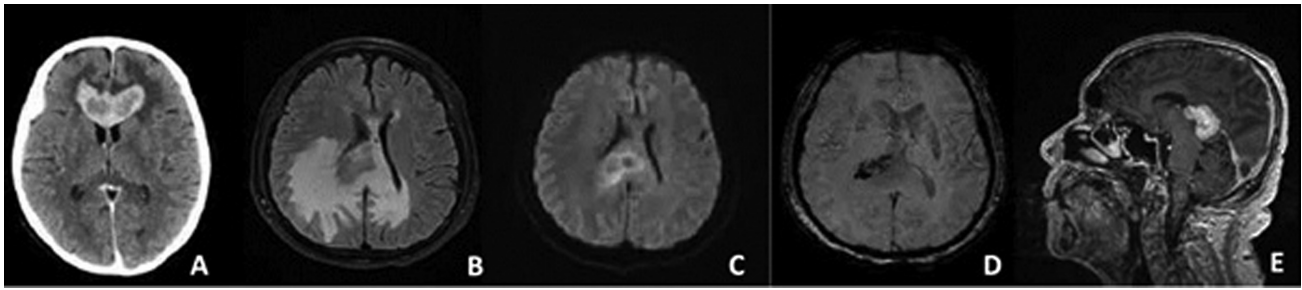


Fig. 4 In a 62-year-old male with HIV-positive status, (A) axial CECT scan shows avidly enhancing mass with the epicenter at the genu of the corpus callosum, crossing the midline involving the both frontal lobes. In another 65-year-old male patient, (B) axial FLAIR, (C) DWI, (D) SWI, and (E) post-contrast T1FS sagittal images show mass with the epicenter in the posterior body and splenium of the corpus callosum in subependymal distribution with the rim of diffusion restriction, areas of hemorrhages, and intense post-contrast enhancement. Diagnosis: HIV-associated primary CNS lymphoma. CECT, contrast-enhanced computed tomography; CNS, central nervous system; DWI, diffusion-weighted imaging; FLAIR, fluid-attenuated inversion recovery; HIV, human immunodeficiency virus; SWI, susceptibility-weighted imaging; T1FS, T1-weighted fat-saturated; T2WI, T2-weighted imaging.

Imaging: HIV-related PCNSL is characterized by lesions with a larger diameter (≥ 4 cm) than CNS-toxoplasmosis, with an almost equal chance of numerous and solitary lesions. They most typically involve the basal ganglia and corpus callosum.¹²

CT scan shows hyperdense lesions with homogeneous post-contrast enhancement, as in ►Fig. 4(A). On MRI, the lesion appears as iso- to hypointense relative on T1WI, mostly hypointense on T2WI (T2-weighted imaging)/FLAIR images, and shows ring enhancement on post-contrast T1WI with areas of hemorrhage and necrosis, shown in ►Fig. 4.

Lower apparent diffusion coefficient value, a greater choline/creatinine ratio on MRS, and a higher regional cerebral blood volume on MR perfusion in individuals with HIV-related PCNSL are effective in distinguishing HIV-related PCNSL from CNS-toxoplasmosis.¹³

Progressive Multifocal Leukoencephalopathy

Pathophysiology: This is caused by the JC virus, leading to demyelination of the white matter.¹⁴ An initial JC virus infection in childhood typically persists in an asymptomatic carrier state in most individuals, and the virus can however reactivate in immunosuppressed patients.

Imaging: Asymmetric, nonenhancing T2/FLAIR hyperintensities, which are hypointense on T1W images, appear in the subcortical and in the periventricular white matter.¹⁵ Decreased NAA and increased Cho and Mi values are seen on MRS due to damage to white matter and destruction of myelin.

PML lesions may be difficult to distinguish from tumefactive conditions like gliomatosis or demyelinating lesions such as multiple sclerosis on MRI. However, typically, the lesions show no mass effect or diffusion restriction on diffusion-weighted imaging (DWI) unlike gliomatosis cerebri. Multiple sclerosis lesions show increased MD and altered FA values due to anisotropy loss. T2WI shows front edge of demyelinating lesions in multifocal leukoencephalopathy appears faintly hyperintense with a large number of discrete, hyperintense spots of demyelination, an appearance similar to that of the Milky Way. The advancing edge of the lesion shows a hyperintense rim on DWI.¹⁴ Another MR sign

diagnostic and typical of PML is the “Shrimp sign” characterized by T2-hyperintensity in the cerebellar white matter abutting the middle cerebellar peduncle, but sparing the dentate nucleus.¹⁶ Features are depicted in ►Fig. 5(A, B).

Cryptococcal Meningitis

Pathophysiology: Cryptococcus is a fungal species that enters the body by inhalation only to be removed by the host's defense mechanisms. But occasionally, particularly in immunocompromised HIV patients, it might result in pneumonia and subsequent CNS spread to cause meningoencephalitis.

Imaging: Meningitis or meningoencephalitis is the major feature of CNS cryptococcosis. Another appearance often seen is when the infection spreads via the perivascular spaces, where gelatinous mucoid cryptococcal capsular polysaccharides and budding yeast build up in the dilated perivascular spaces, forming tiny gelatinous pseudocysts, as shown in ►Fig. 6A(a). These have a typical “soap bubble appearance” on MRI, with intermediate signal on T1W images, high signal on T2WI, and suppression on FLAIR images.¹⁷ These gelatinous pseudocysts are commonly seen in the basal ganglia, thalamus, and midbrain. When there is immunological response from ART administration, imaging may show cryptococcoma and nodular meningeal enhancement similar to what is observed in granulomatous disorders like tuberculosis, sarcoidosis, etc. as shown in ►Fig. 6B(b-d)-⁴.

Tuberculous Meningitis

Pathophysiology: Hematogenous spread of *M. tuberculosis* in an immunocompromised HIV patient results in involvement of meninges or cerebral parenchyma and tubercular abscess formation.⁴

Imaging: Although imaging findings in HIV and non-HIV-associated CNS-TB are essentially identical, meningitis is more common in persons infected with HIV. Inflammation of small blood vessels, e.g., lenticulostriate branches, causes vasospasm, resulting in the infarction predominantly seen in bilateral basal ganglia.

Classic imaging findings in patients with tubercular meningitis (TBM) include communicating hydrocephalus (75%), basilar exudates (38%), periventricular infarctions (15–30%),

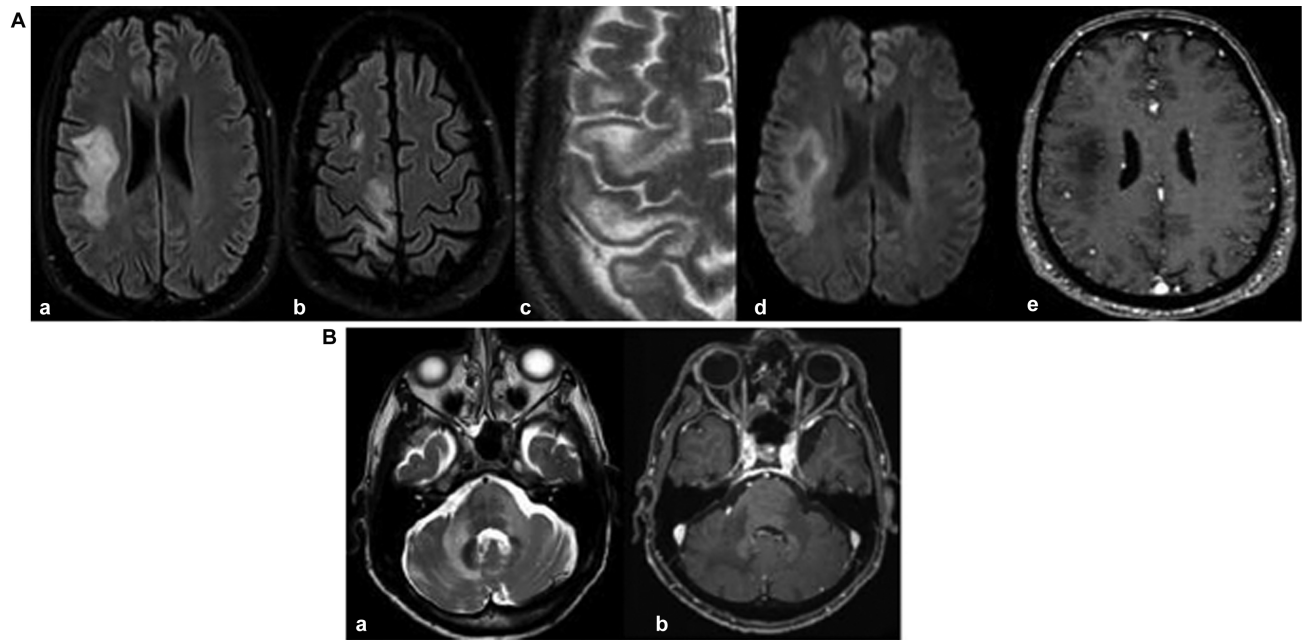


Fig. 5 (A) In a 54-year-old male with HIV-positive status, (a, b) axial FLAIR images show asymmetric hyperintensity in subcortical white matter of right the fronto-parietal region (involving subcortical U fibers) with extension into deep white matter, and (c) Milky Way, axial T2WI image shows T2WI multiple tiny round hyperintense dots at advancing edge. (d) Axial DWI images show diffusion restriction of the leading edge with a central hypointense signal and (e) post-contrast T1FS axial images show no obvious enhancement. Diagnosis: Progressive multifocal leukoencephalopathy. (B) In a 40-year-old male with HIV-positive status, shrimp sign (a) T2WI images show hyperintensity in bilateral middle cerebellar peduncle and cerebellar white matter with sparing of dentate nucleus and (b) post-contrast T1FS axial images show no enhancement. Diagnosis: Progressive multifocal leukoencephalopathy. CNS, central nervous system; DWI, diffusion-weighted imaging; FLAIR, fluid-attenuated inversion recovery; HIV, human immunodeficiency virus; T1FS, T1-weighted fat-saturated; T2WI, T2-weighted imaging.

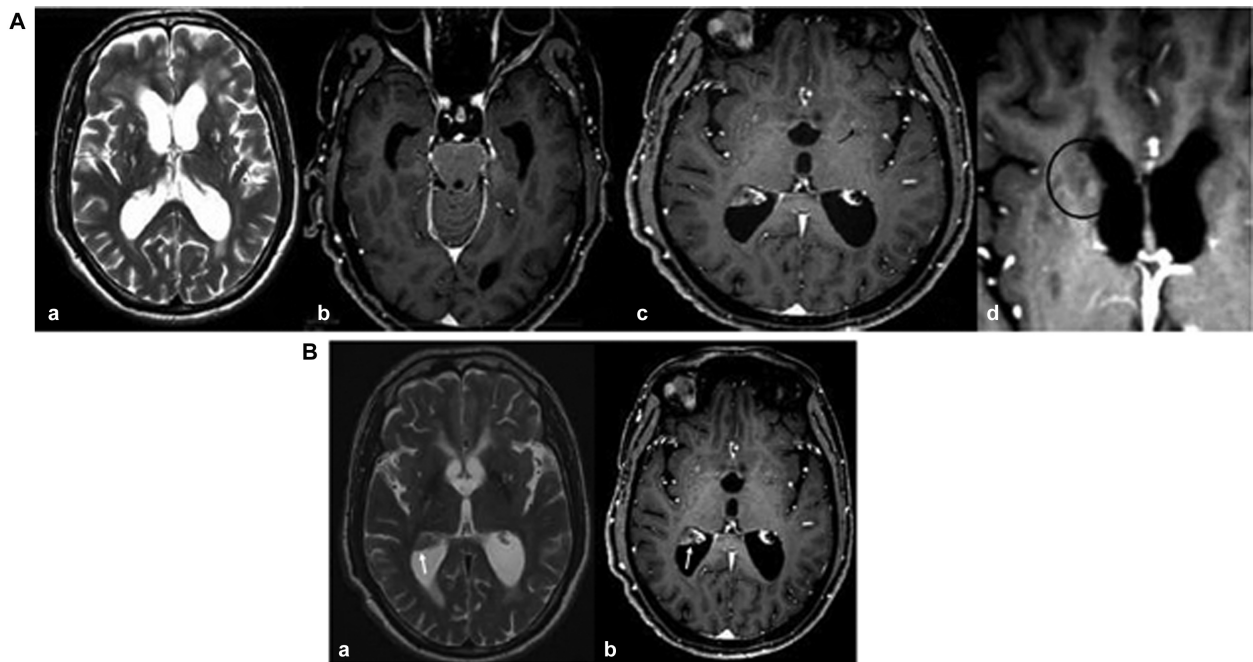


Fig. 6 (A) In a 50-year-old male PLHIV patient with cryptococcal Ab-positive, (a) axial T2WI images show dilated perivascular spaces in bilateral basal ganglia with ventriculomegaly. (b) Axial post-contrast T1FS images show meningeal enhancement due to exudates in the basal cisterns and along the brainstem (c, d). Axial post-contrast T1FS images show small ring-enhancing cysts in basal ganglia in the left putamen and right caudate nucleus (circle). Diagnosis: Cryptococcal meningitis. (B) In a 50-year-old male PLHIV patient with cryptococcal Ab-positive, choroid plexitis is observed. There is asymmetric enlargement of choroid plexus of posterior horn of right lateral ventricle, appearing T2 iso-hypointense, in figure (a), with nodular heterogeneous post-contrast enhancement on T1WI images in figure (b). PLHIV, person living with HIV; T1FS, T1-weighted fat-saturated; T1WI, T1-weighted imaging; T2WI, T2-weighted imaging.

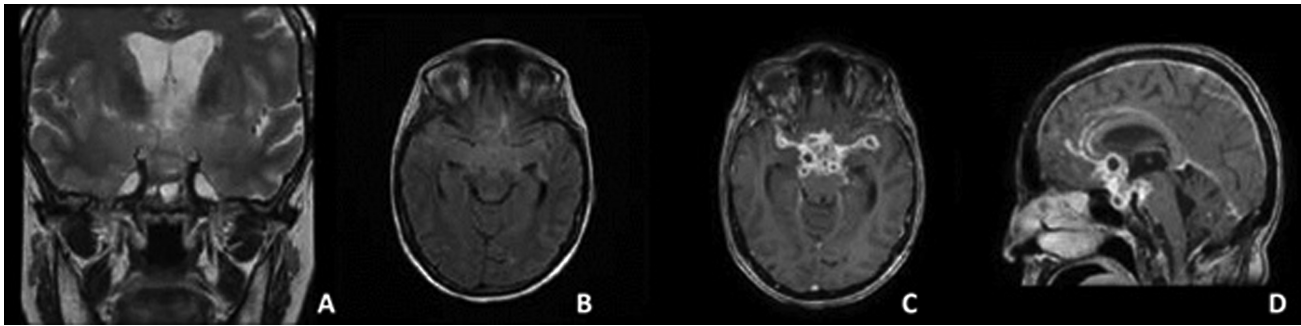


Fig. 7 In a 7-year-old male PLHIV patient of TB meningitis, (A) coronal T2 and (B) FLAIR images show hyperintense lesions in basal cisterns with hypointense rim. (C) Axial post-contrast T1FS and (D) sagittal post-contrast T1FS images show multiple ring-enhancing lesions in basal cistern with extensively thickened enhancing meninges. **Diagnosis: Tubercular meningitis.** PLHIV, person Living with HIV; T1FS, T1-weighted fat-saturated; T1WI, T1-weighted imaging; TB, tuberculosis.

and cerebral parenchymal tuberculomas (5–10%).¹⁸ Caseating tuberculomas shows a T2WI hypointense rim, with restriction on DWI and ring enhancement on post-contrast T1WI, while noncaseating tuberculoma shows homogenous nodular enhancement.⁴ Imaging detailed in ►Fig. 7.

Immune Reconstitution Inflammatory Syndrome

Pathophysiology: IRIS is an exuberant inflammatory response to an underlying antigen. In patients of HIV, a variable onset is noted, ranging from weeks to years, after the initia-

tion of highly active ART. Those with baseline CD4 count <50 cells are at increased risk. Patient usually presents with worsening of the existing symptoms or development of new symptoms despite adequate therapy. In most cases, this inflammatory response is mild and self-limiting; however, in some individuals a fulminant manifestation can be observed, causing rapid worsening of patient's symptoms and eventually death shortly after onset. With such uncertain course, and incidence reported to be as high as 25 to 35%, its importance cannot be undermined.¹⁹

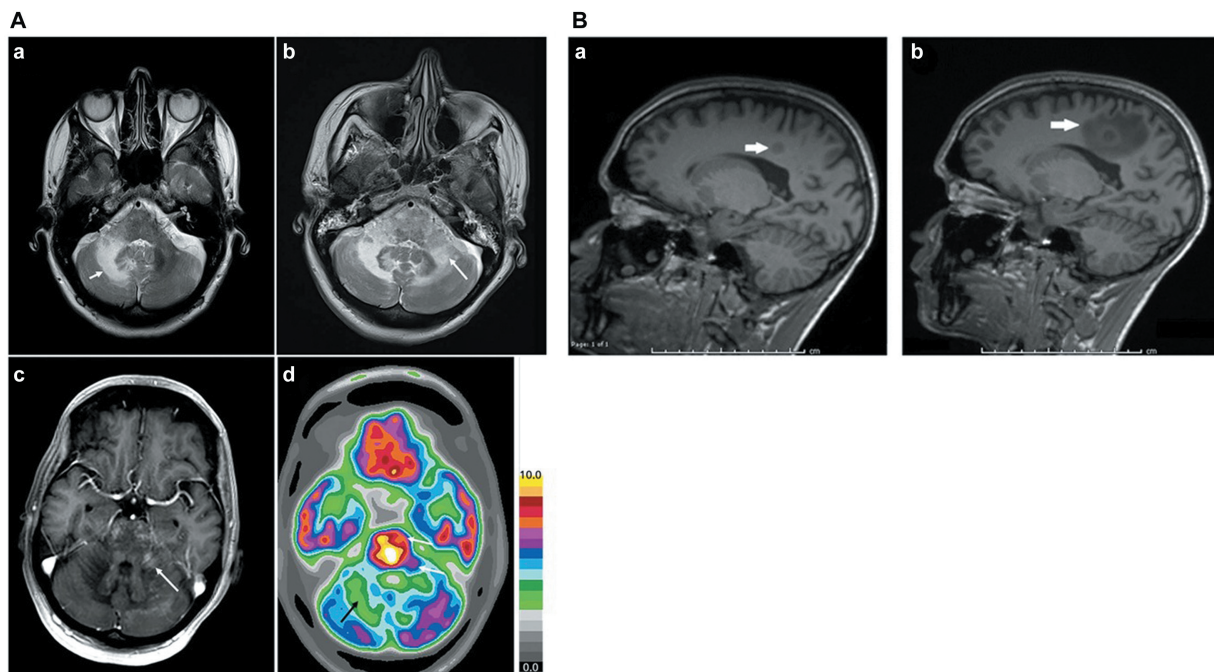


Fig. 8 (A) (a, arrow) T2-weighted image shows right peridentate and middle cerebellar peduncle (MCP) hyperintensity consistent with progressive multifocal leukoencephalopathy (PML). Follow-up examination revealed florid T2 hyperintensity in the cerebellum and brainstem (b, arrow), with enhancement (c). FDG-PET shows right cerebellar hypometabolism (d, black arrow) at original PML site and hypermetabolism (d, white arrows) in pons and left MCP consistent with PML-immune reconstitution inflammatory syndrome. (Image reproduced with permission of B. Aravindhnan et al; doi:10.1212/WNL.0000000000006213.)²¹ (B) MRI of the brain with contrast, with T1 MP-RAGE sequences shown. (a) MRI of the brain obtained at the time of initial TB diagnosis, showing a 1.0×1.2 cm area of enhancement in the right mesial parietal lobe. (b) MRI of the brain obtained at the time of re-presentation and diagnosis with CNS TB-IRIS. The previously seen area of enhancement is slightly increased in size to 1.0×1.3 cm, and there is new surrounding vasogenic edema, measuring 4.5×2.3 cm. (Image reproduced with permission of C. Elizabeth et al; <https://doi.org/10.1093/ofid/ofac367>.)²² CNS, central nervous system; FDG-PET, fluorodeoxyglucose positron emission tomography; MRI, magnetic resonance image; TB, tuberculosis; TB-IRIS, immune reconstitution inflammatory syndrome to tuberculosis.

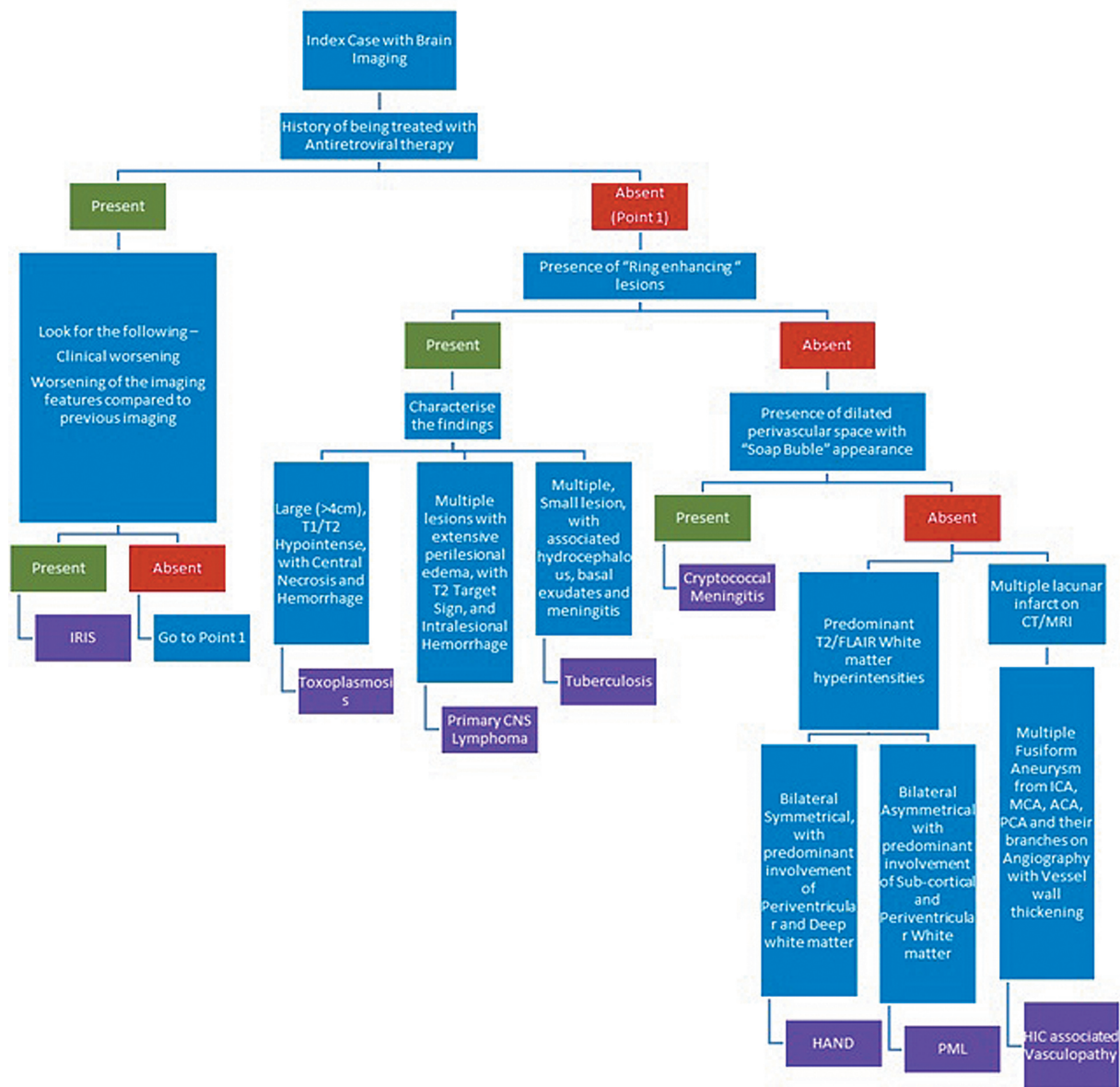


Fig. 9 Flowchart showing a systematic pattern-based approach to intracranial central nervous system (CNS)-HIV complications. IRIS, immune reconstitution inflammatory syndrome; HAND, HIV-associated neurocognitive disorder; PML, progressive multifocal leukoencephalopathy.

Imaging: diverse clinical presentation and imaging manifestation of CNS-IRIS make it difficult to diagnose. MRI shows an increase in the parenchymal abnormalities on FLAIR, mass effect, and diffusion restriction. On post-contrast scan, enhancement of parenchymal abnormality usually points toward CNS-IRIS, in appropriate clinical settings.¹⁹

► **Figs. 8(A, B) and 9** present representative cases of CNS-IRIS, PML-IRIS, and TB-IRIS respectively, reproduced with permission of the respective authors.

Pattern-Based Approach

Radiological imaging is pivotal in the diagnosis, differentiation, and management of intracranial complications associated with HIV. Each condition presents with unique imaging characteristics which, when correctly interpreted, can significantly influence clinical outcomes.

Thus, we provide a pattern-based approach to CNS pathologies in HIV patients in ► **Table 2**.

Authors' Contributions

Conceptualization: D.V., S.M., and L.B.
 Formal analysis and investigation: D.V., S.M., and L.B.
 Methodology: D.V., S.M., and L.B.
 Validation: D.V., S.M., and L.B.
 Writing—original draft: D.V., S.M., and L.B.
 Writing—review and editing: D.V., S.M., and L.B.
 Supervision: D.V., S.M., and L.B.

Ethical Approval

Ethical approval was waived by the local Ethics Committee of ABVIMS and Dr. RML Hospital, New Delhi, in view of the retrospective nature of the study and all the procedures being performed were part of the routine care.

Table 2 Pattern-based approach to intracranial CNS pathologies in HIV patients

Primary pattern	1		2		
	Predominant T2/FLAIR white matter hyperintensities		Ring-enhancing lesions		
Secondary pattern	Bilateral symmetric involvement	Bilateral asymmetric involvement	Multiple lesions with extensive perilesional edema, with T2 target sign, and intralesional hemorrhage	Large (>4 cm), T1/T2 hypointense, with central necrosis and hemorrhage	Multiple, small lesion, with associated hydrocephalous, basal exudates, and meningitis
Additional information	Predominant involvement of periventricular and deep white matter	Predominant involvement of subcortical and periventricular white matter	Areas Involved: thalamus, cerebellum, subcortical white and basal ganglia	Areas involved: basal ganglia, corpus callosum	Areas Involved: basal cisterns and cerebral sulcal spaces
Diagnosis	HAND	PML	Toxoplasmosis	Primary CNS lymphoma	Tuberculosis
Primary pattern	3		4	5	
	Meningitis		Multiple infarct on NCCT	Posttreatment flare-up	
Secondary pattern	Dilated perivascular spaces with soap bubble appearance, predominantly in basal ganglia, thalamus, and midbrain	Multiple ring-enhancing lesions, with basal exudates and hydrocephalus	Multiple fusiform aneurysm from ICA, MCA, ACA, PCA, and their branches on angiography	T2/FLAIR hyperintensities, diffusion restriction, and patchy enhancement.	
Additional information	Choroid plexitis	Caseating and noncaseating nodules	Vessel wall thickening on vessel wall imaging	Interval resolution of changes is noted.	
Diagnosis	Cryptococcal meningitis.	Tuberculosis	HIV associated vasculopathy	IRIS	

Abbreviations: ACA, anterior cerebral artery; CNS, central nervous system; FLAIR, fluid-attenuated inversion recovery; HAND, HIV-associated neurocognitive disorder; HIV, human immunodeficiency virus; IRIS, immune reconstitution inflammatory syndrome; MCA, middle cerebral artery; MIP, maximum intensity projection; PCA, posterior cerebral artery; PML, progressive multifocal leukoencephalopathy.

Patients' Consent

Informed consent was obtained from all individual participants included in the study.

Conflict of Interest

None declared.

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